WireShark Lab #1 - Getting Started

학과: 컴퓨터소프트웨어학부

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1. Which of the following protocols are shown as appearing (i.e., are listed in the Wireshark "protocol" column) in your trace file: TCP, QUIC, HTTP, DNS, UDP, TLSv1.2?

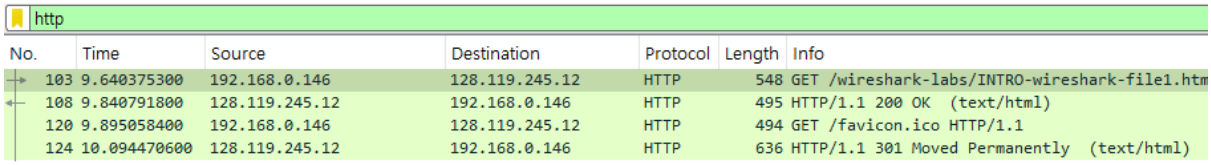
TLSv1.3, DNS, TCP, HTTP, UDP

2. How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received? (By default, the value of the Time column in the packetlisting window is the amount of time, in seconds, since Wireshark tracing began. (If you want to display the Time field in time-of-day format, select the Wireshark View pull down menu, then select Time Display Format, then select Time-of-day.)

GET 패킷 전송: 9.640s

200 ok 패킷 수신: 9.840s

따라서 OK 수신까지 걸린 시간은 0.200s 이다.



http

No.	Time	Source	Destination	Protocol	Length	Info
103	9.640375300	192.168.0.146	128.119.245.12	HTTP	548	GET /wireshark-labs/INTRO-wireshark-file1.htm
108	9.840791800	128.119.245.12	192.168.0.146	HTTP	495	HTTP/1.1 200 OK (text/html)
120	9.895058400	192.168.0.146	128.119.245.12	HTTP	494	GET /favicon.ico HTTP/1.1
124	10.094470600	128.119.245.12	192.168.0.146	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

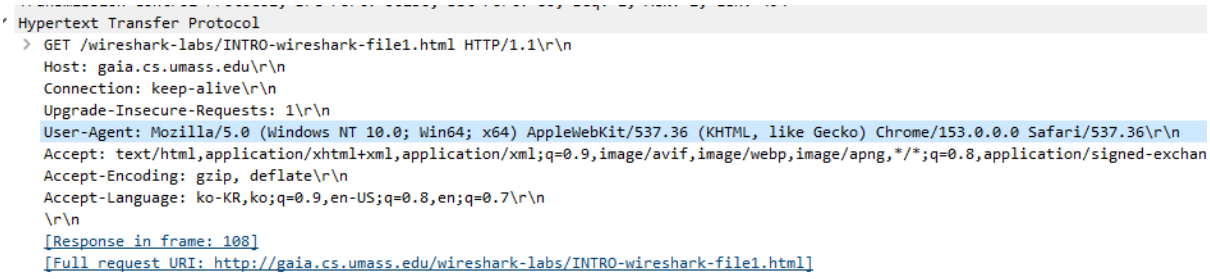
3. What is the Internet address of the gaia.cs.umass.edu (also known as wwwnet.cs.umass.edu)? What is the Internet address of your computer or (if you are using the trace file) the computer that sent the HTTP GET message?

gaia.cs.umass.edu: 128.119.245.12

local: 192.168.0.146

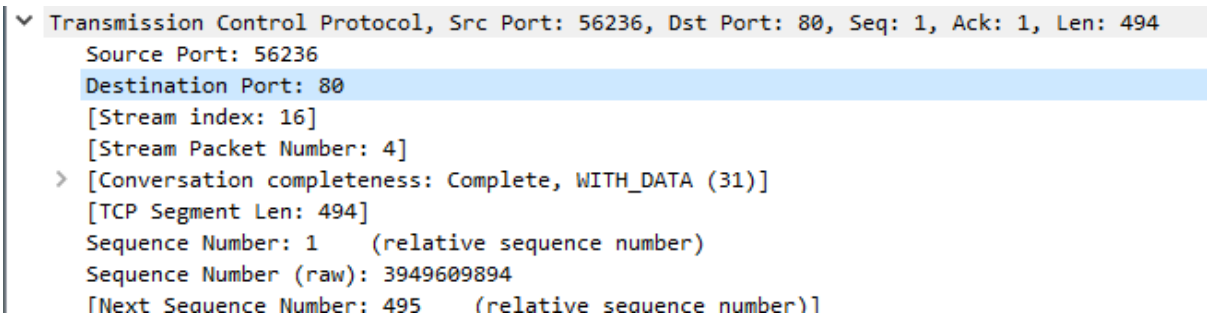
4. Expand the information on the HTTP message in the Wireshark "Details of selected packet" window (see Figure 3 above) so you can see the fields in the HTTP GET request message. What type of Web browser issued the HTTP request? The answer is shown at the right end of the information following the "User-Agent:" field in the expanded HTTP message display. [This field value in the HTTP message is how a web server learns what type of browser you are using.] • Firefox, Safari, Microsoft Internet Edge, Other

Chrome을 사용하였다



5. Expand the information on the Transmission Control Protocol for this packet in the Wireshark "Details of selected packet" window (see Figure 3 in the lab writeup) so you can see the fields in the TCP segment carrying the HTTP message. What is the destination port number (the number following "Dest Port:" for the TCP segment containing the HTTP request) to which this HTTP request is being sent?

Destination Port: 80



6. Print the two HTTP messages (GET and OK) referred to in question 2 above. To do so, select Print from the Wireshark File command menu, and select the "Selected Packet Only" and "Print as displayed" radial buttons, and then click OK.

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C:\Users\splab\AppData\Local\Temp\wireshark_Wi-FiSX7YV3.pcapng 전체 패킷 216개, 2개 표시됨

No.     Time           Source                Destination           Protocol Length Info
    103 9.640375300    192.168.0.146         128.119.245.12        HTTP     548    GET /wireshark-labs/INTRO-wireshark-
file1.html HTTP/1.1
Frame 103: Packet, 548 bytes on wire (4384 bits), 548 bytes captured (4384 bits) on interface \Device\NPF_{916FA072-
FCE5-4FB7-9F55-2315BD7E2468}, id 0
Ethernet II, Src: EFMNetworks_ff:ca:db (58:86:94:ff:ca:db), Dst: EFMNetworks_bd:0c:e4 (90:9f:33:bd:0c:e4)
Internet Protocol Version 4, Src: 192.168.0.146, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 56236, Dst Port: 80, Seq: 1, Ack: 1, Len: 494
    Source Port: 56236
    Destination Port: 80
    [Stream index: 16]
    [Stream Packet Number: 4]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 494]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3949609894
    [Next Sequence Number: 495    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3146398422
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
    Window: 513
    [Calculated window size: 131328]
    [Window size scaling factor: 256]
    Checksum: 0x0aeb [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (494 bytes)
Hypertext Transfer Protocol
No.     Time           Source                Destination           Protocol Length Info
    108 9.840791800    128.119.245.12        192.168.0.146         HTTP     495    HTTP/1.1 200 OK  (text/html)
Frame 108: Packet, 495 bytes on wire (3960 bits), 495 bytes captured (3960 bits) on interface \Device\NPF_{916FA072-
FCE5-4FB7-9F55-2315BD7E2468}, id 0
Ethernet II, Src: EFMNetworks_bd:0c:e4 (90:9f:33:bd:0c:e4), Dst: EFMNetworks_ff:ca:db (58:86:94:ff:ca:db)
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.0.146
Transmission Control Protocol, Src Port: 80, Dst Port: 56236, Seq: 1, Ack: 495, Len: 441
    Source Port: 80
    Destination Port: 56236
    [Stream index: 16]
    [Stream Packet Number: 6]
    [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 441]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3146398422
    [Next Sequence Number: 442    (relative sequence number)]
    Acknowledgment Number: 495    (relative ack number)
    Acknowledgment number (raw): 3949610388
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
    Window: 501
    [Calculated window size: 64128]
    [Window size scaling factor: 128]
    Checksum: 0x8b02 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
    [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (441 bytes)
Hypertext Transfer Protocol
Line-based text data: text/html (3 lines)
    <html>\n
    Congratulations!  You've downloaded the first Wireshark lab file!\n
    </html>\n
```